Global Power Plant Database
- Analysis

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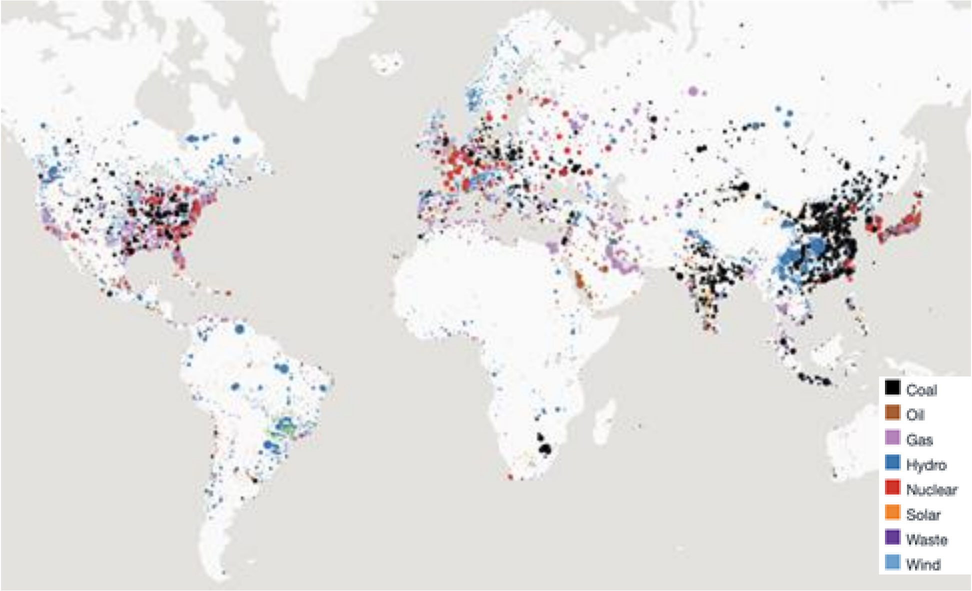
Agenda

1. **Executive Summary**
2. **Business Use Cases**
3. **Methodology**
4. **Data Analysis**
5. **Visualizations**
6. **Recommendations**
7. **Lessons Learned**

Data Overview

- **34,937 records** of **164 countries**.
  - **72%** of the world's electricity generation capacity
- Each record contains:
  - **Power plant details** (2013 to 2017):
    - geolocations
    - energy source
    - plant capacity
    - yearly actual and estimated energy generation
    - ownership
- **Data Quality**:
  - **> 90% of plants in the US** reported energy generation data.
  - **< 50% of plants for other countries** reported their energy generation data

Power Plants by Capacity (MW) and Fuel Type



Data Source: Global Power Plant Database

Executive Summary

Observation:

The global energy sector is undergoing a significant transformation: **increasing focus on renewable energy sources and sustainable practices**.

Key challenges:

- **Uneven adoption** of renewables across regions
- **Compliance** with emission standards
- **Economic** impact on economies.

Problem Statement:

Energy stakeholders lack actionable insights to identify renewable energy investment opportunities, ensure emission compliance, and quantify the economic contributions of power plants.

Goal:

To provide data-driven insights that support renewable energy investment, emission compliance, and economic impact analysis for informed decision-making by energy stakeholders.

Business Use Cases

Identifying Renewable Energy Hotspots:

Identify regions leading in adding renewable energy sources over the past few years

➔ **Goal: Help policymakers** identify thriving regions to replicate successful strategies in other areas and allocate resources more effectively.

Emission Compliance Monitoring:

Identify CO_2 emissions for each energy fuel type

➔ **Goal: Identify energy fuel types** that contribute the most to CO_2 emissions to strategize on transitioning to cleaner, low-emission alternatives.

Regional Economic Impact by Fuel Type:

Analyze the economic contribution of power plants by fuel type in terms of employment.

➔ **Goal**: **Identify** the fuel types generating the most employment

Data Processing Tools

I. Data Cleaning

- Python
- MS-Excel

I. Data Modeling and Analysis

- MySQL
- MySQL Workbench
- Google Cloud

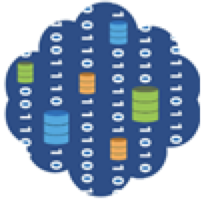
III. Data Visualization

- Tableau

IV. Reporting and Recommendations

- MS-PowerPoint

Extract | Transform | Load



Collect raw data:

1. **Primary Dataset**

[Global Power Plant Database]

1. **Secondary Datasets**:
   a. Employment data
   [US Bureau Labor Statistics]
   a. CO2 Emissions data
   [EPA]

1. **Clean Data:**
- Trim white spaces
- Remove missing values
- Remove duplicates and non-ASCII data

1. **Normalization:**
- Transform to **2NF**
- Transform to **3NF**

→ Removal of partial and transitive dependency

3. **Export Tables** to separate CSV files for each table

Load Google Cloud Database with data

Normalization: Concept Model

Step 1: Data was initially in First Normal Form (1NF) :

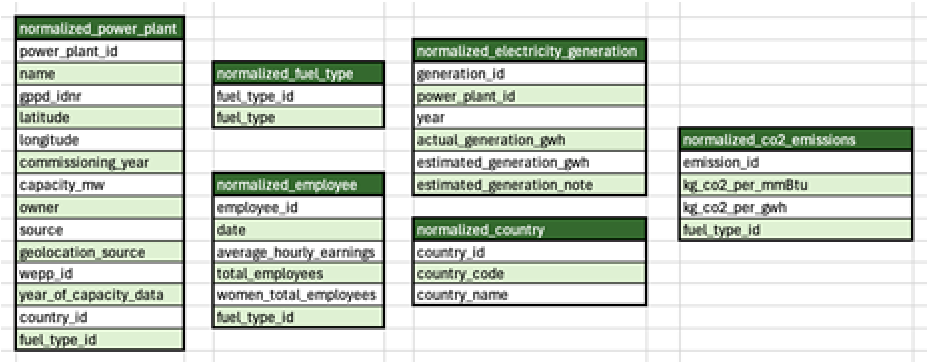
- Ensured atomic values
- Ensure no repeating groups

Step 2: Transformation to Second Normal Form (2NF) :

- Removal of partial functional dependencies

Step 3: Transformation to Third Normal Form (3NF) :

- Removal of transitive functional dependencies



Data Transformation

Step 1: Load Original Datasets into Python

- **Load** data in raw form

Step 2: Create Separate Data Frames for Each Normalized Tables

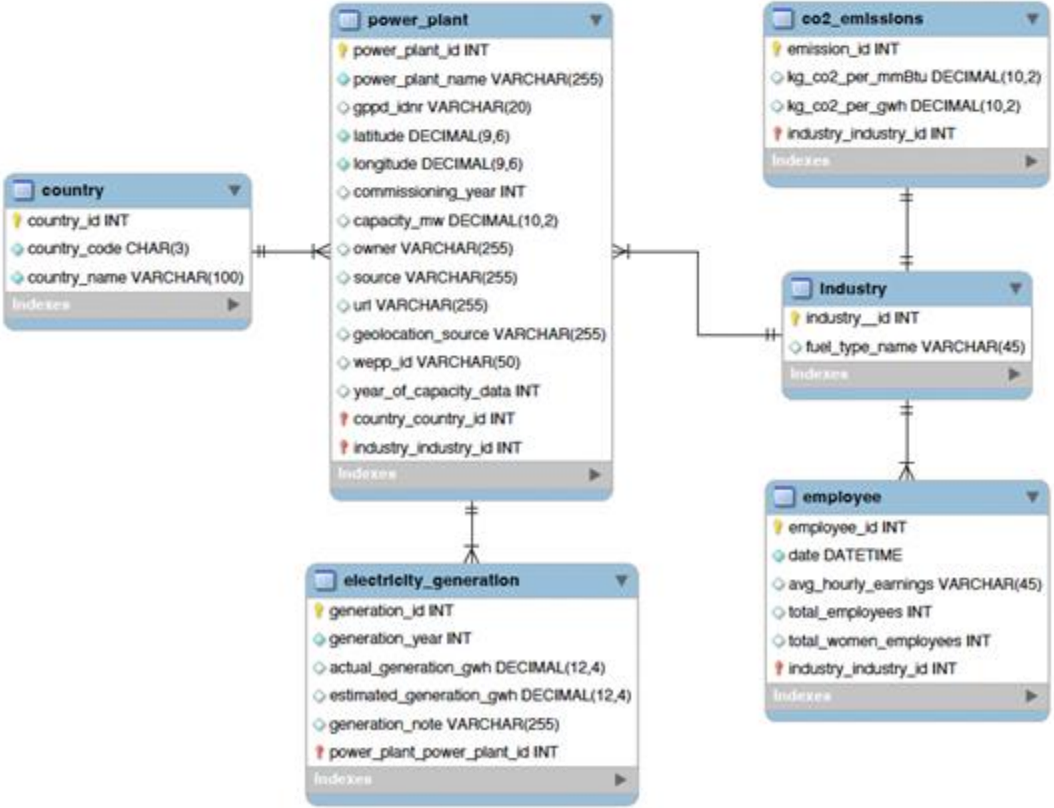
- **Select** columns from relevant datasets
- **Ensured correct** data types
- **Cleaned** the data by removing missing values, duplicates and non-ASCII data.

Step 3: Export data frames into separate .CSV files:

- normalized_co2_emissions.csv
- normalized_country.csv
- normalized_electricity_generation.csv
- normalized_employee.csv
- normalized_fuel_type.csv
- normalized_power_plant.csv

Relational Modeling: EER Diagram

I. **Created** tables and columns on MySQL Workbench
II. **Established** Primary Keys and Foreign Key **Constraints**
III. **Create Relationships** between tables
IV. **Forward Engineered** to generate table creation code (Data Definition Language)
V. Final Output: **Enhanced Entity Relationship (EER) Diagram**



SQL Query 1:
Identifying Renewable Energy Hotspots

➔ **Help policymakers identify thriving regions** to replicate successful strategies in other areas and allocate resources more effectively.

Code:

```sql
SELECT
    c.country_name,
    p.name AS power_plant_name,
    ft.fuel_type,
    p.capacity_mw,
    p.latitude,
    p.longitude
FROM
    power_plant p
JOIN
    fuel_type ft ON p.fuel_type_id = ft.fuel_type_id
JOIN
    country c ON p.country_id = c.country_id
WHERE
    ft.fuel_type IN ('solar', 'wind', 'hydro')
ORDER BY
    p.capacity_mw DESC;
```

Output:

country_name	power_plant_name	fuel_type	capacity_mw
China	Three Gorges D...	hydro	22500.0000
China	Baihetan Dam	hydro	13050.0000
China	Xiluodu	hydro	12600.0000
Venezuela	Simon Bolivar (...	hydro	8851.0000
Brazil	Tucuru	hydro	8535.0000
Brazil	Itaipu (Parte Br...	hydro	7000.0000
Paraguay	Itaipu Binacion...	hydro	7000.0000

Report Query 1:
Identifying Renewable Energy Hotspots

Global Renewable Energy Capacity by Country and Energy Fuel Type

Top 3 Hotspot Countries per Energy Source:

- **Hydro:** China, Venezuela and Brazil
- **Solar:** China, India, and United States
- **Wind:** China, United Kingdom and United States

Key:

- ➔ **Highlighted countries** have renewable energy capacities ranging from **200 - 225,000 MW**
- ➔ **Labeled countries** exceed energy capacity of **3,000 Mw**

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SQL Query 2:
Emissions Compliance Monitoring

➔ **Identify energy fuel types** that contribute the most to CO_2 emissions to strategize on transitioning from high-emission fuels to cleaner, low-emission alternatives.

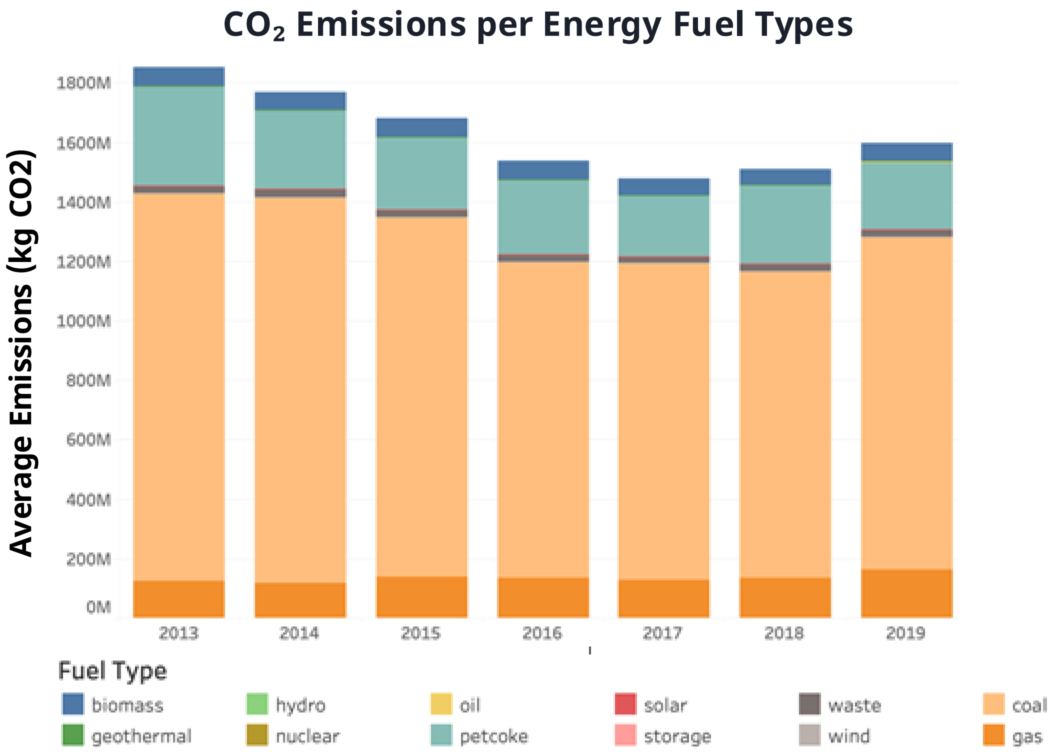
Code:

```
SELECT
    pp.power_plant_id,
    c.country_code,
    eg.generation_year,
    ft.fuel_type,
    ROUND(ce.kg_CO2_per_gwh * eg.actual_generation_gwh, 4) AS emmisions
FROM
  co2_emissions ce
    JOIN
  fuel_type ft ON ft.fuel_type_id = ce.fuel_type_id
        JOIN
    power_plant pp ON pp.fuel_type_id = ft.fuel_type_id
    JOIN
    country c ON c.country_id = pp.country_id
    JOIN
  electricity_generation eg ON pp.power_plant_id = eg.power_plant_id
WHERE
  c.country_code = 'USA'
    AND ROUND(ce.kg_CO2_per_gwh * eg.actual_generation_gwh, 4) != 0
ORDER BY
  emmisions DESC;
```

Output:

power_plant_id	country_code	generation_year	fuel_type	emmisions
32392	USA	2014	coal	6331680766.3822
29054	USA	2019	coal	6196432546.0227
28080	USA	2017	coal	6030826680.0470
28080	USA	2014	coal	5947252784.8191
28080	USA	2018	coal	5908527032.9816
28080	USA	2013	coal	5807880847.2329
29431	USA	2013	coal	5795566864.6824

Report Query 2:
Emissions Compliance Monitoring

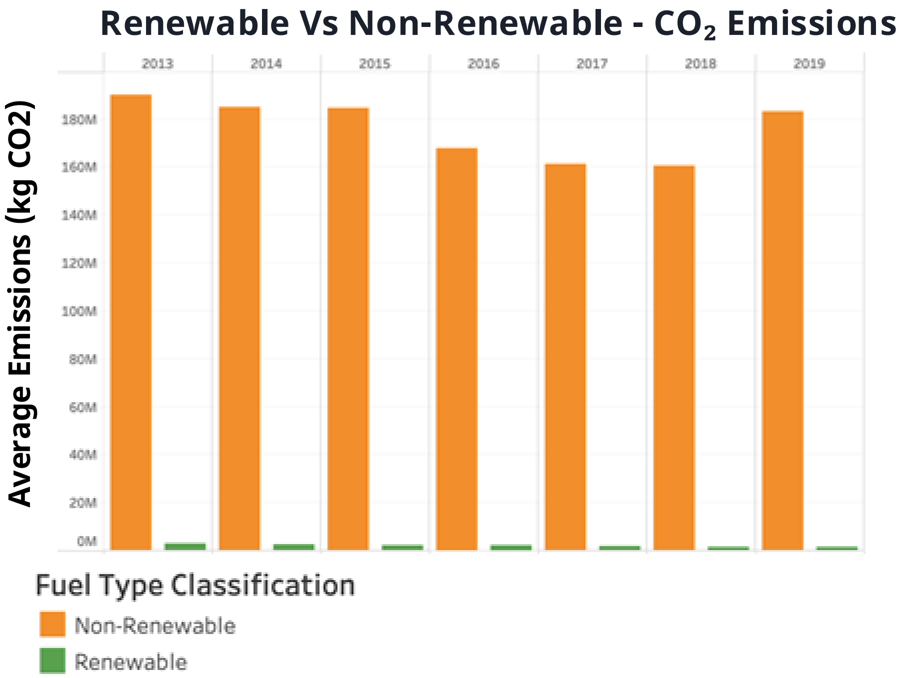


Breakdown of CO_2 emissions per fuel type.

Insights:

- **Petcoke** and **Coal** remain the largest contributors to emissions.
- **Overall decrease** in emissions throughout the years.

Report Query 2:
Emissions Compliance Monitoring



Breakdown of CO_2 emissions of renewable vs. non-renewables.

Insights:

- **Stark contrast** between renewable and non-renewable CO2 emissions.
- **Decrease** between non-renewables from 2016-2018 but sharp increase in 2019.

SQL Query 3:
Economic Impact by Fuel Type

- ➔ **Analyze the economic contribution** of power plants by fuel type in terms of employment.
- ➔ **Identify** the fuel types generating the most employment

Code:

```sql
SELECT
  YEAR(e.date) AS year,
  SUM(e.total_employees) AS total_employees_per_fuel_type,
  ft.fuel_type
FROM
  employee e
JOIN
  fuel_type ft
ON
  ft.fuel_type_id = e.fuel_type_id
GROUP BY
  year,
  ft.fuel_type
ORDER BY
  year DESC;
```

Output:

year	total_employees_per_fuel_type	fuel_type
2012	77	biomass
2012	207	coal
2012	262	gas
2012	77	geothermal
2012	77	hydro
2012	77	nuclear
2012	207	oil

Report Query 3:

Economic Impact by Fuel Type

Yearly Total Employee Per Energy Fuel Type

Fuel Type	2014	2015	2016	2017	2018	2019	2020
biomass	125	132	132	132	132	131	126
coal	391	384	381	365	355	334	320
gas	449	456	453	450	440	444	443
geothermal	125	132	132	132	132	131	126
hydro	125	132	132	132	132	131	126
nuclear	125	132	132	132	132	131	126
oil	391	384	381	365	355	334	320
petcoke	391	384	381	365	355	334	320
solar	125	132	132	132	132	131	126
waste	193	193	198	204	211	215	209
wind	125	132	132	132	132	131	126

Top Employment Trends:

- **Gas** consistently employs **2-3 times more** workers than other fuel types.
- **Oil**, **coal**, and **petcoke** have seen an **18.15% decline** in employees over the past 6 years.
- **Employment in all renewable** energy fuel types has remained **stagnant** over the same period.

SQL Query 4:
Economic Impact by Fuel Type - Gender Gap

- ➔ **Examine women representation** in total employees across different energy fuel types
- ➔ **Highlight gender disparities** and promote discussions on improving women's representation in the sector.

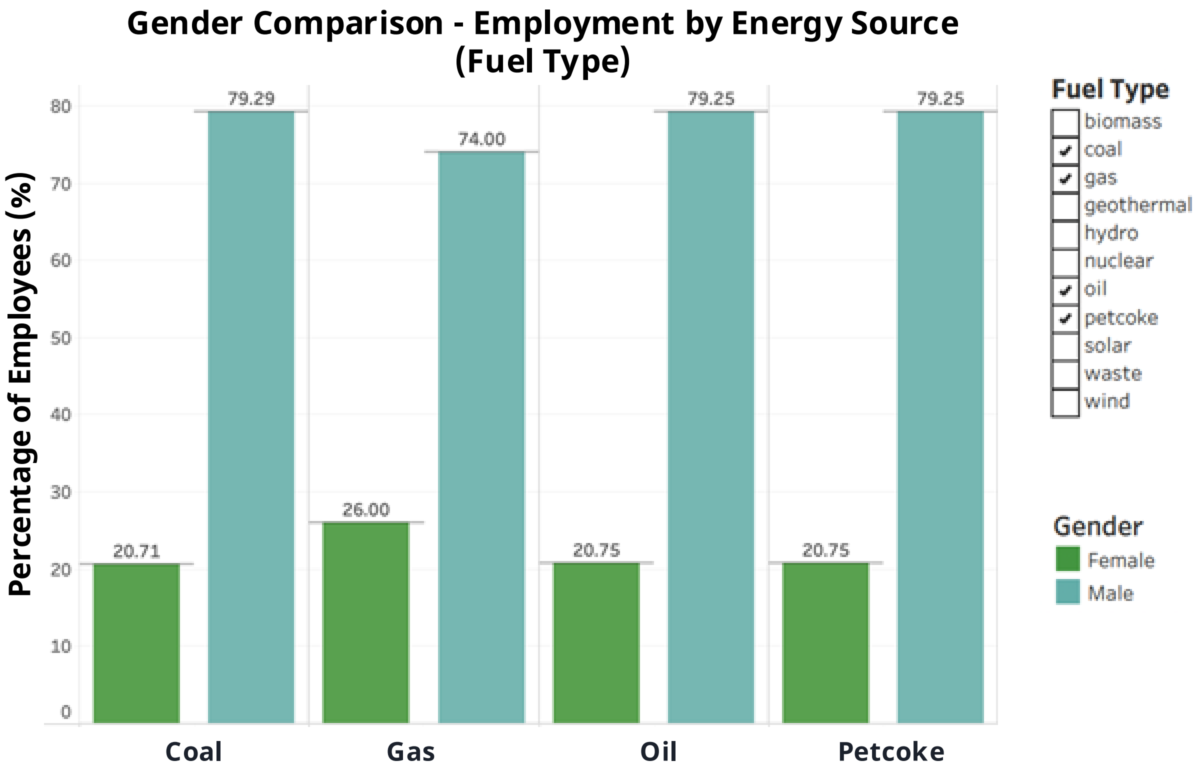
Code:

```sql
SELECT
  SUM(e.total_women_employees) AS total_women_by_fuel_type,
  (SUM(e.total_women_employees) / SUM(e.total_employees)) * 100
  AS percentage_female_employees,
  100-(SUM(e.total_women_employees) / SUM(e.total_employees)) * 100
  AS percentage_male_employees,
  f.fuel_type
FROM
  employee e
INNER JOIN
  fuel_type f
ON
  e.fuel_type_id = f.fuel_type_id
GROUP BY
  e.fuel_type_id,
  f.fuel_type
ORDER BY
  total_women_by_fuel_type DESC;
```

Output:

total_women_by_fuel_type	percentage_female_employees	percentage_male_employees	fuel_type
815	25.9968	74.0032	gas
525	20.7510	79.2490	oil
525	20.7510	79.2490	petcoke
524	20.7115	79.2885	coal
370	26.0014	73.9986	waste
0	0.0000	100.0000	biomass
0	0.0000	100.0000	geothermal

Report Query 4:
Economic Impact for Non-Renewable Energy - Gender Gap



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Insights:

- **Significant gender gap**, with men dominating the field, hinting at systemic issues in recruiting or retaining female employees.
- **In coal, oil, and petcoke sectors,** female employment is **below 21%**, indicating significant gender disparity.
- Near-equal distribution in the aggregate of all power plants.

Dashboard
Employment Trend

Top Insights:

- **Gas** consistently employs **2-3 times more** workers than other fuel types.
- **Significant gender gap**, with men dominating the field

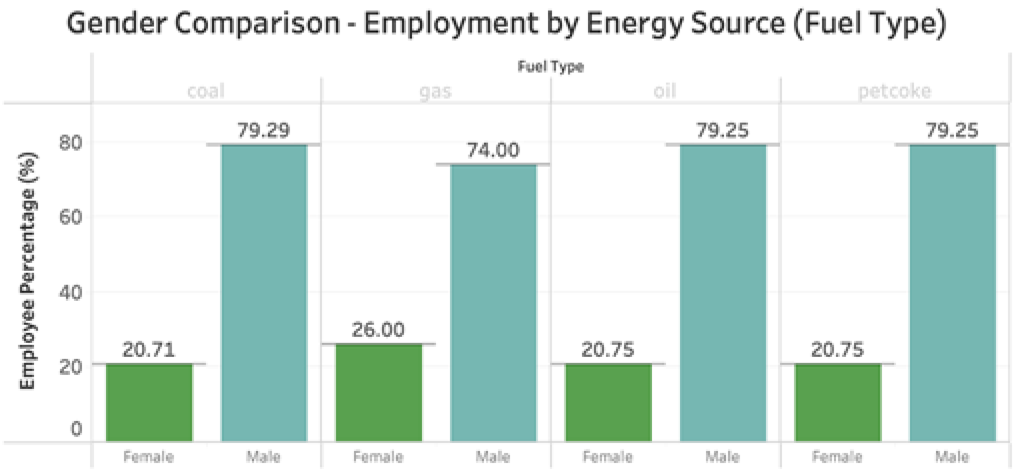
Yearly Total Employee Per Energy Fuel Type

Fuel Type	2014	2015	2016	2017	2018	2019	2020
biomass	125	132	132	132	132	131	126
coal	391	384	381	365	355	334	320
gas	449	456	453	450	440	444	443
geothermal	125	132	132	132	132	131	126
hydro	125	132	132	132	132	131	126
nuclear	125	132	132	132	132	131	126
oil	391	384	381	365	355	334	320
petcoke	391	384	381	365	355	334	320
solar	125	132	132	132	132	131	126
waste	193	193	198	204	211	215	209
wind	125	132	132	132	132	131	126

Fuel Type: biomass, coal ✓, gas ✓, geothermal, hydro, nuclear, oil ✓, petcoke ✓, solar, waste, wind

Gender: Female, Male

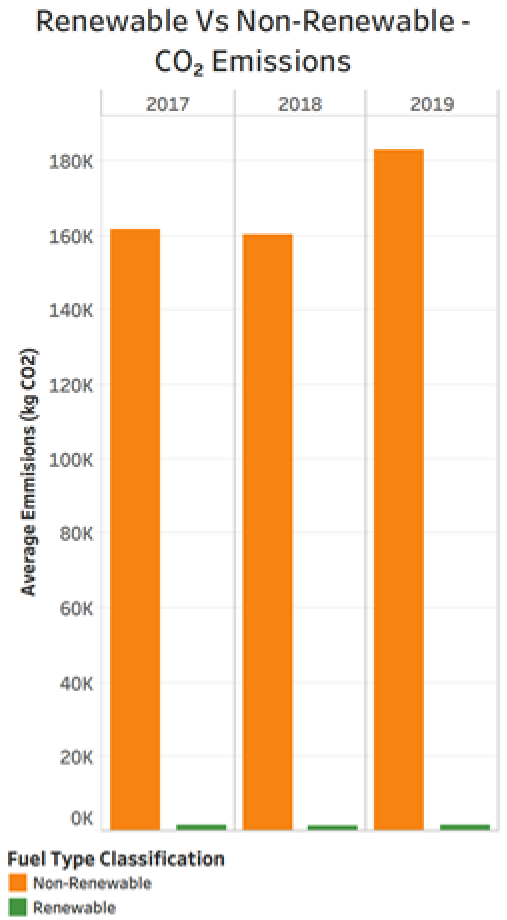
Total Employees: 125 – 456

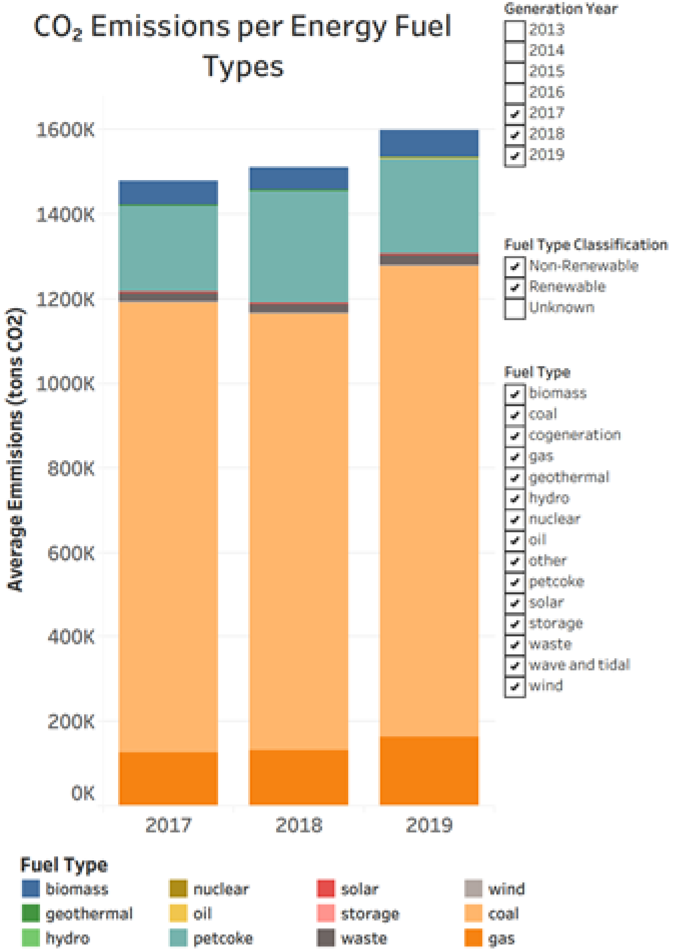


Dashboard
CO_2 Emission Trend

Top Insights:

- Non-renewable Fuels significantly dominate the CO_2 emissions.
- Coal and Gas contribute the majority of the CO2 emissions





Recommendations

Identifying Renewable Energy Hotspots:

- ➔ Identified hotspot countries indicate potential opportunities for further investment
- ➔ Hotspot countries can assist untapped countries in implementing renewable energy sources

Energy Generation Performance Monitoring:

- ➔ Improve Forecasting Accuracy
- ➔ Enhance Operational Efficiency
- ➔ Develop Real-Time Monitoring Systems

Regional Economic Impact by Fuel Type:

- ➔ Expand workforce training and development in renewable energy sectors.
- ➔ Boost investment and innovation to attract talent to renewables.
- ➔ Strengthen industry-education partnerships to build a skilled workforce.

Economic Impact by Fuel Type - Gender Gap:

- ➔ Boost women workforce in non renewable fuel type power plants.
- ➔ Promote Gender Diversity in Renewable Energy.
- ➔ Awareness Campaigns to Reduce Stereotypes.

Lessons Learned

1. Women are still **under represented** in employment
2. Providing Context: **Paris Agreement (Treaty) 2016**: United States and China main contributors.
    a. United States joined in 2016, exited 2017-2020 and rejoined in 2021 (might withdraw in 2025). State-level initiatives also may impact USA's contribution.
    b. China is an emerging leader during United States’s shifting positions
3. **Hydro is still most popular** within renewable energy sources for efficiency and cost.
4. **Gas remains dominant in the employment sector**, highlighting its continued importance in the energy mix.
5. Employment data suggests that there has **not been an increase** in the number of **plants generating renewable energy**.
6. **Stagnation in renewables** emphasizes the importance of innovation and strategic investment to drive growth.

Questions?